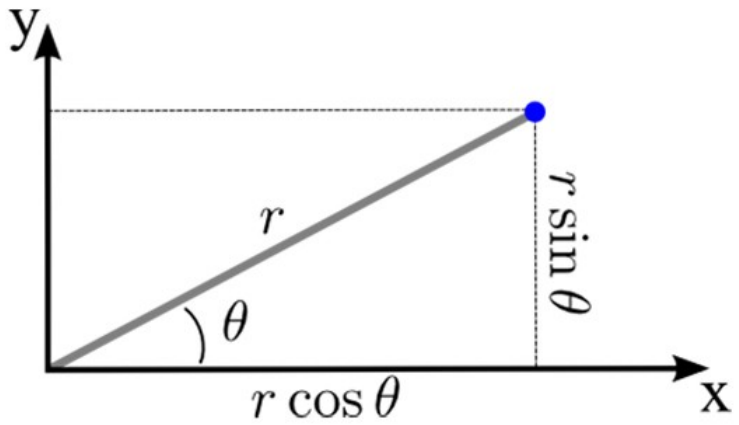
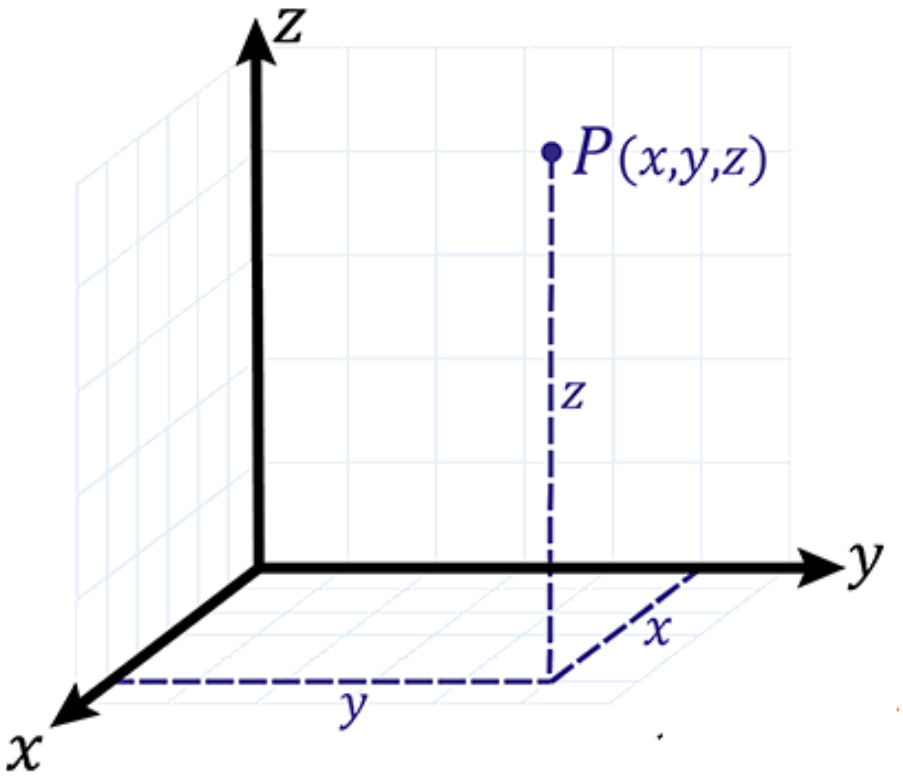
Module 2: Mathematical Physics I

Frame of refrence

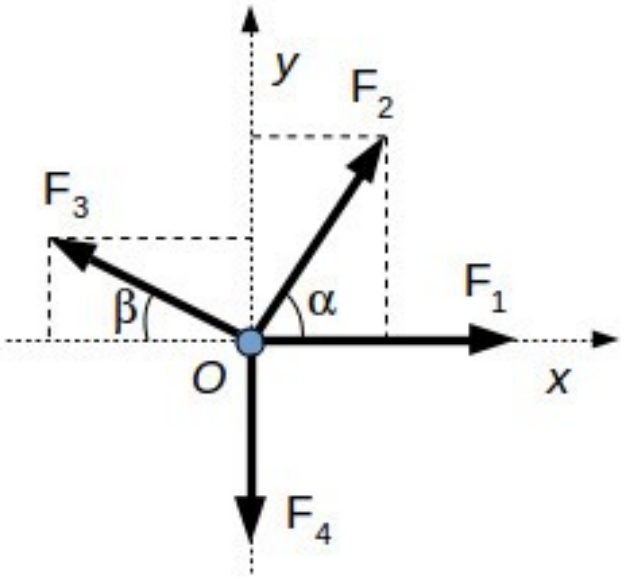


Three- Dimensional Cartesian Coordinate

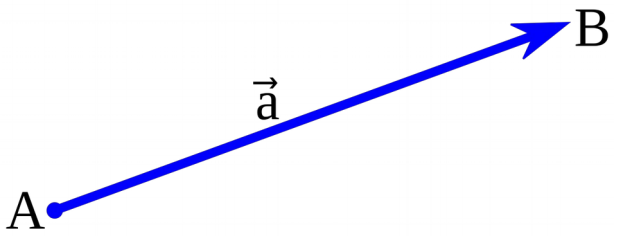


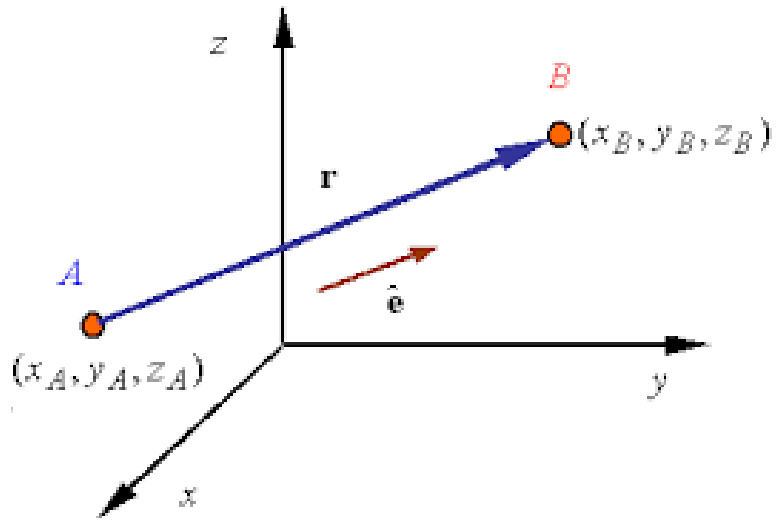
Scalar and vector

Scalar	Vector
Scalar quantity has only magnitude, but no direction	Vector quantity has both magnitude and direction.
Scalar quantities follow the rules of ordinary algebra.	Vector quantity follows the rules of vector algebra.
Divison of scalar with scalar is permitted.	Division between the vectors is not permitted.
Example: Mass, length, and energy etc.	Example: Force, displacement, and weight etc.



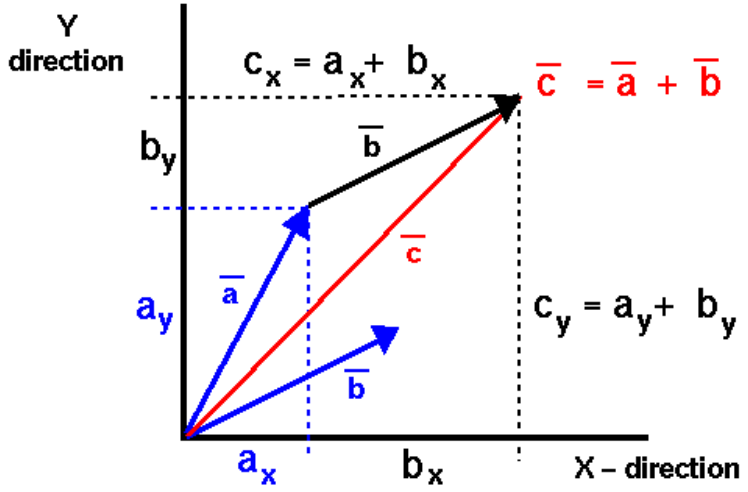
Vectors opreations



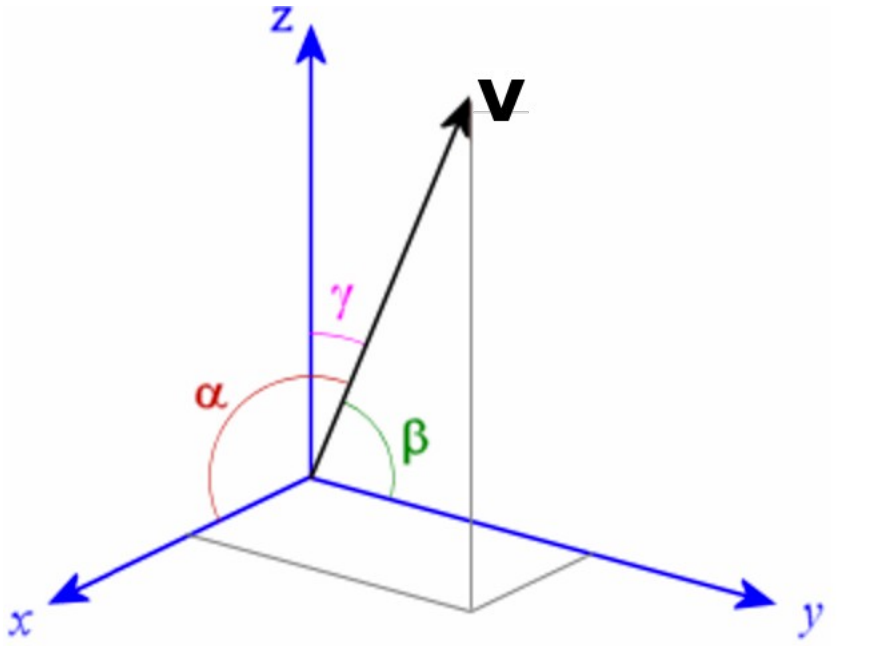


A vector quantity has both magnitude and direction.

Add the vector components.



Vectors opreations



Unit vector $\dfrac{\mathbf{v}}{|\mathbf{v}|}$

$$\alpha = \cos a = \frac{\mathbf{v}\cdot\mathbf{e}_x}{\|\mathbf{v}\|} = \frac{v_x}{\sqrt{v_x^2+v_y^2+v_z^2}},$$

$$\beta = \cos b = \frac{\mathbf{v}\cdot\mathbf{e}_y}{\|\mathbf{v}\|} = \frac{v_y}{\sqrt{v_x^2+v_y^2+v_z^2}},$$

$$\gamma = \cos c = \frac{\mathbf{v}\cdot\mathbf{e}_z}{\|\mathbf{v}\|} = \frac{v_z}{\sqrt{v_x^2+v_y^2+v_z^2}}.$$

Vectors operations

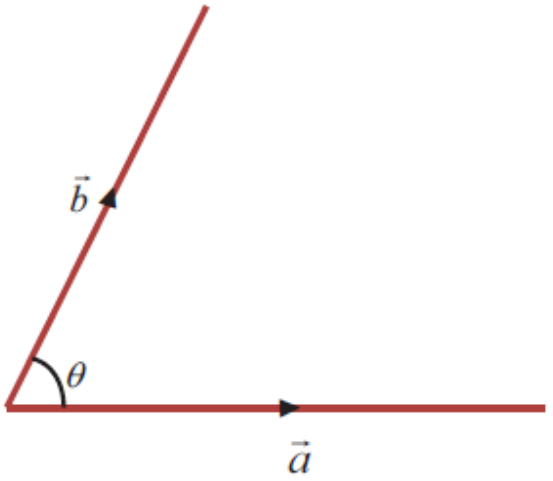
$$\vec{A}\cdot\vec{B} = A_xB_x + A_yB_y + A_zB_z \quad \text{where} \quad \vec{A} = A_x\vec{i} + A_y\vec{j} + A_z\vec{k}, \quad \vec{B} = B_x\vec{i} + B_y\vec{j} + B_z\vec{k}$$

The **dot product** of $\mathbf{u} = \langle u_1, u_2 \rangle$ and $\mathbf{v} = \langle v_1, v_2 \rangle$ is given by $\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2$

Properties of the Dot Product

Let **u, v** and **w** be vectors and let *c* be a scalar.

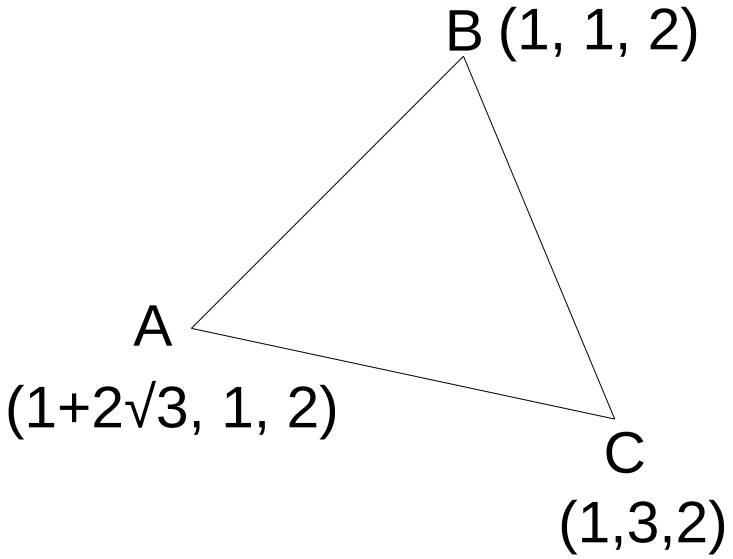
1. $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2. $\mathbf{0} \cdot \mathbf{v} = 0$
3. $\mathbf{u} \cdot (\mathbf{v} + \mathbf{w}) = \mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{w}$
4. $\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2$
5. $c(\mathbf{u} \cdot \mathbf{v}) = c\mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot c\mathbf{v}$



$$\vec{a}\cdot\vec{b} = |\vec{a}|\,|\vec{b}|\cos\theta$$

Vectors operations

Find the angle between side AC and side AB of a triangle ABC. Coordinates of the vertices A,B,C are (1+2√3, 1, 2), (1, 1, 2), and (1,3,2) respectively.



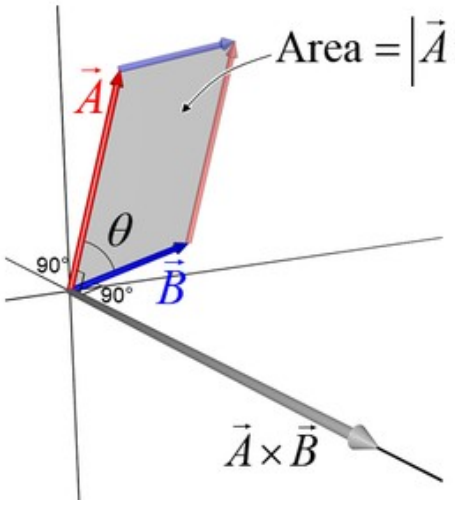
Vectors operations

$$\vec{a} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k} \;, \quad \vec{b} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$$

$$\vec{a}\times\vec{b} = (a_1\hat{i} + a_2\hat{j} + a_3\hat{k}) \times (b_1\hat{i} + b_2\hat{j} + b_3\hat{k})$$

$$\vec{a}\times\vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

The cross product is all about area and calculating vectors that are perpendicular to $\vec{A}$ and $\vec{B}$.



$$\vec{A}\times\vec{B} = |\vec{A}||\vec{B}|\sin\theta\hat{a}_n$$

$\hat{a}_n$ is a unit vector perpendicular to the plane defined by $\vec{A}$ and $\vec{B}$.

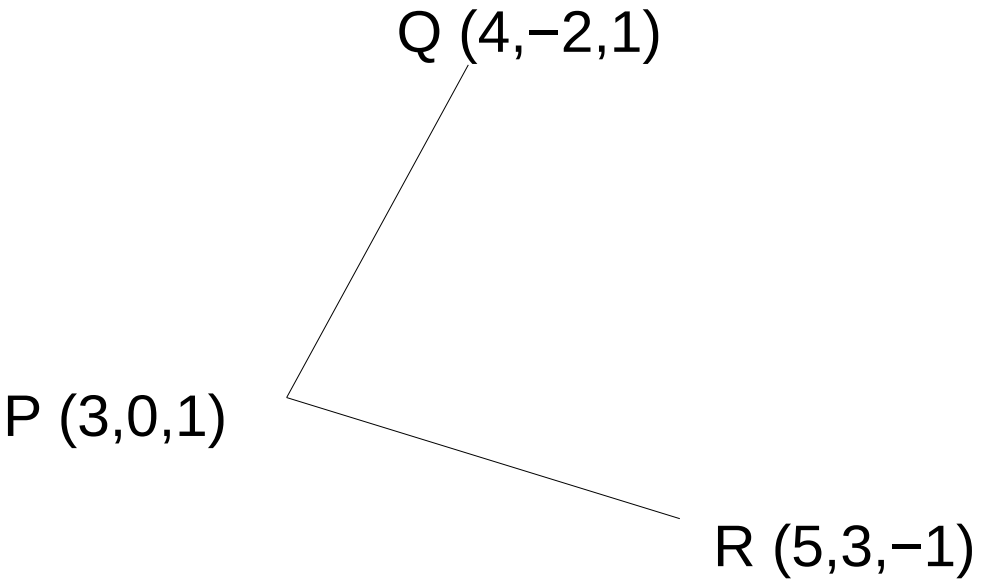
Vectors operations

If $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are vectors and c is a scalar, then

1. $\mathbf{a} \times \mathbf{b} = -(\mathbf{b} \times \mathbf{a})$
2. $(c\mathbf{a}) \times \mathbf{b} = c(\mathbf{a} \times \mathbf{b}) = \mathbf{a} \times (c\mathbf{b})$
3. $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$
4. $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c}$
5. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$
6. $\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}$

Vectors operations

Find area vector and area that is orthogonal to the plane containing the points P=(3,0,1), Q=(4,−2,1), and R=(5,3,−1).



Gradient, divergence and curl

$$\nabla = \frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k$$

$$\nabla f = \left[i \frac{\partial}{\partial x} + j \frac{\partial}{\partial y} + k \frac{\partial}{\partial z} \right] f$$

f is scalar

$$\boldsymbol{F} = F_1 \boldsymbol{i} + F_2 \boldsymbol{j} + F_3 \boldsymbol{k} = \left\langle F_1, F_2, F_3 \right\rangle$$

$$\nabla = \frac{\partial}{\partial x} \boldsymbol{i} + \frac{\partial}{\partial y} \boldsymbol{j} + \frac{\partial}{\partial z} \boldsymbol{k} = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$$

$$\text{Div} = \nabla \cdot \boldsymbol{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

$$\text{Curl} = \nabla \times \boldsymbol{F} = \begin{vmatrix} \boldsymbol{i} & \boldsymbol{j} & \boldsymbol{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$$

Gradient, divergence and curl

$\varphi = x^3yz^3$ and $\mathbf{A} = xz\,\mathbf{i} - y^2\,\mathbf{j} + 2x^2y\,\mathbf{k}$

Evaluate

(a) $\nabla\varphi$

(b) $\nabla . \mathbf{A}$

(c) $\nabla \mathrm{x}\ \mathbf{A}$

Integral

$$\boldsymbol{F} = F_x \boldsymbol{i} + F_y \boldsymbol{j} + F_z \boldsymbol{k}$$

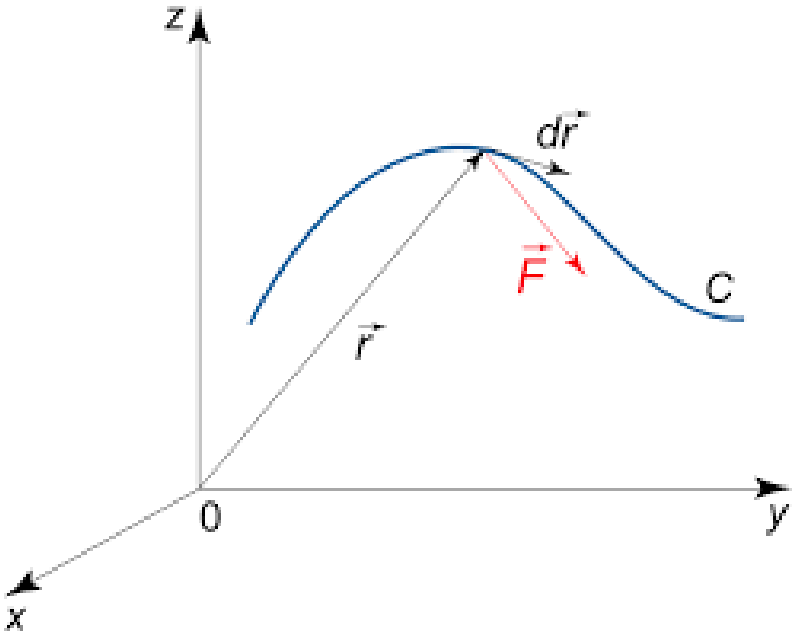
$$d\boldsymbol{r} = dx\,\boldsymbol{i} + dy\,\boldsymbol{j} + dz\,\boldsymbol{k}$$

$$\int_c \boldsymbol{F} . d\boldsymbol{r} = \int_c F_x dx + F_y dy + F_z dz$$

For **F** = $(2xy+z^3)$**i** + (x^2+2y)**j** + $(3xz^2-2)$**k,**

Calculate $\int_c \boldsymbol{F} . d\boldsymbol{r}$

C is the path joining (1,-1,1), and (2,1,2)



Matrix

$$3a - b = 7$$
$$-a + 7b - 6c = 2$$
$$-3b + 4c = 0$$

$$\begin{pmatrix} 3 & -1 & 0 \\ -1 & 7 & -6 \\ 0 & -3 & 4 \end{pmatrix} \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} 7 \\ 2 \\ 0 \end{pmatrix}$$

Rectangular list of numbers (or functions).

$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ \vdots \\ m \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & \dots & n \end{array} \\ \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ a_{31} & a_{32} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \end{array}$$

Matrix

$$A = [a_{ij}] = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \square & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}_{m \times n}$$

column matrix

$$\begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

row matrix

$$\begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$$

$$B = \begin{pmatrix} 5 & 6 \\ 8 & 7 \end{pmatrix}$$

square matrix

$$\begin{pmatrix} 2 & 5 & 7 & 8 \\ 1 & 2 & 3 & 1 \\ 4 & 5 & 0 & 1 \end{pmatrix}$$

rectangular matrix

Diagonal matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Symmetric matrix

$$\begin{bmatrix} 1 & 2 & 6 \\ 2 & 8 & 4 \\ 6 & 4 & 5 \end{bmatrix}$$

Matrix operations

$$\mathbf{A}+\mathbf{B}=\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}+\begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \cdots & b_{mn} \end{bmatrix}$$

$$=\begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} & \cdots & a_{1n}+b_{1n} \\ a_{21}+b_{21} & a_{22}+b_{22} & \cdots & a_{2n}+b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1}+b_{m1} & a_{m2}+b_{m2} & \cdots & a_{mn}+b_{mn} \end{bmatrix}$$

$$\begin{bmatrix} 4 & 8 \\ 3 & 7 \end{bmatrix}+\begin{bmatrix} 1 & 0 \\ 5 & 2 \end{bmatrix}=\begin{bmatrix} 4+1 & 8+0 \\ 3+5 & 7+2 \end{bmatrix}$$

Subtraction

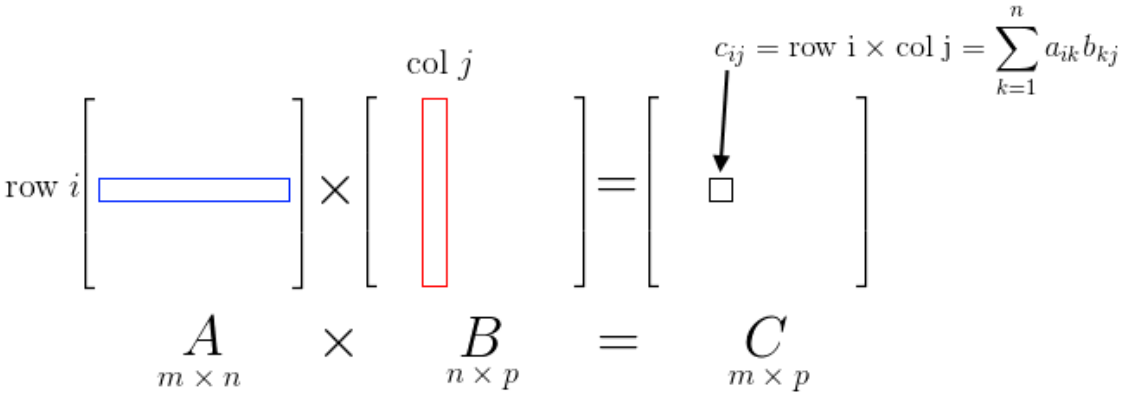
$$\begin{bmatrix} A11 & A12 \\ A21 & A22 \end{bmatrix}-\begin{bmatrix} B11 & B12 \\ B21 & B22 \end{bmatrix}=\begin{bmatrix} A11-B11 & A12-B12 \\ A21-B21 & A22-B22 \end{bmatrix}$$

$$\mathbf{A}=\begin{bmatrix} 0 & 1 \\ 4 & 2 \end{bmatrix},\quad \mathbf{B}=\begin{bmatrix} 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$\mathbf{A}-\mathbf{B}=\begin{bmatrix} 0 & 1 \\ 4 & 2 \end{bmatrix}-\begin{bmatrix} 2 & 5 \\ 3 & 6 \end{bmatrix}=\begin{bmatrix} 0-2 & 1-5 \\ 4-3 & 2-6 \end{bmatrix}$$

$$\mathbf{A}-\mathbf{B}=\begin{bmatrix} -2 & -4 \\ 1 & -4 \end{bmatrix}$$

Matrix operations



$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix} \begin{bmatrix} b_1 & b_2 & b_3 \\ b_4 & b_5 & b_6 \\ b_7 & b_8 & b_9 \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \\ c_7 & c_8 & c_9 \end{bmatrix}$$

Matrix operations

$$A = \begin{pmatrix} 1 & -2 & 4 & 5 \\ 7 & -8 & -6 & 2 \\ 2 & 3 & -1 & -2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} -2 & 1 & -3 \\ 0 & -2 & -5 \\ 4 & -3 & -7 \\ 0 & 0 & 6 \end{pmatrix}$$

Multiply the two matrices?

Matrix operations

$$2 \cdot \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} = \begin{bmatrix} 2.1 & 2.2 & 2.3 \\ 2.4 & 2.5 & 2.6 \\ 2.7 & 2.8 & 2.9 \end{bmatrix}$$

Multiplication by a scalar

$$= \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \\ 14 & 16 & 18 \end{bmatrix}$$

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

Division by a scalar

$$\frac{A}{2} = \begin{pmatrix} \frac{1}{2} & \frac{2}{2} \\ \frac{3}{2} & \frac{4}{2} \end{pmatrix} = \begin{pmatrix} 0.5 & 1 \\ 1.5 & 2 \end{pmatrix}$$

Matrix operations

$$I_n = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{pmatrix}$$ Identity Matrix

For any given matrices A, B and C of the order *m* x *n*

A + B = B + A

A + (B + C) = (A + B) + C

k (A + B) = *k*A + *k*B, here *k* is a scalar

Matrix operations

$A = \{a_{ij}\}_{mxn}$, $B = \{b_{ij}\}_{nxp}$, and $C = \{c_{ij}\}_{pxk}$

$(A\,B)\,C = A\,(B\,C)$

$A\,(B + C) = A\,B + A\,C$ or $(A + B)\,C = A\,C + BC$

$I\,A = A\,I = A$

$$A = \begin{pmatrix} 1 & 1 & -1 \\ 2 & 0 & 3 \\ 3 & -1 & 2 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 3 & 1 \\ 0 & 2 & -1 \\ -1 & 4 & 2 \end{pmatrix} \qquad C = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & -2 \\ -4 & 1 & 2 \end{pmatrix}$$

Evaluate $A^3 - 4A^2 - 3A$ $\qquad A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{pmatrix}$

Matrix operations

$A = \{a_{ij}\}_{mxn}$ then

A' or $A^T = \{a_{ji}\}_{nxm}$ is called the transpose of A

For any sutable matrices A and B

$(A')' = A$

$k\,(A)' = (k\,A)'$ where k is a scalar

$(A + B)' = A' + B'$

$(AB)' = B'\,A'$

$A' =$

3	4	4
3	2	1
2	0	2

$$A = \begin{pmatrix} 3 & 3 & 2 \\ 4 & 2 & 0 \\ 4 & 1 & 2 \end{pmatrix} \qquad B = \begin{pmatrix} 2 & -1 & 2 \\ 1 & 2 & 4 \\ 3 & 6 & 1 \end{pmatrix}$$

Prove the above identities

Also calculate
$(A + B)\,(A - B)$

Matrix operations

Any matrix A = $\{a_{ij}\}_{mxm}$ is said to symmetric if A = A'. If A = -A', then matrix is said to be skew, symmetric.

If a matrix is skew symmetric, then $a_{ii} = -a_{ii} = 0$, i.e., all the diagonal elements of a skew symmetric matrix are 0.

For any square matrix A, (A + A') is symmetric and (A – A') is skew symmetic.

Let B = (A + A'), then B' = (A + A')'

B' = A' + (A')' = (A' + A) = (A + A') = B

Matrix operations

C = (A – A’) then C’ = (A – A’)’

C’ = A’ - (A’)’ = (A’ - A) = - (A - A’) = -C

Thus, any symmetric matrix can be expressed as sum of a symmetric and a skew symmetric matrix.

A = 1/2(B + C) = 1/2 (A + A’) + 1/2 (A - A’)

Prove the previous identites for

$$A = \begin{pmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{pmatrix}$$

Matrix operations

Matrix transformation

(1) The interchange of any two rows or two columns.

$$R_2 \leftrightarrow R_3 \quad \begin{bmatrix} 1 & -1 & 2 & 0 \\ -2 & 3 & 7 & 1 \\ 4 & 5 & -6 & -3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -1 & 2 & 0 \\ 4 & 5 & -6 & -3 \\ -2 & 3 & 7 & 1 \end{bmatrix}$$

(2) The multiplication of the elements of any row or column by a non zero number.

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{array}\right] (3) \sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 15 & 18 & 21 & 24 \\ 9 & 10 & 11 & 12 \end{array}\right]$$

Matrix operations

(3) The addition to the elements of any row or column, the corresponding elements of any other row or column multiplied by any non zero number.

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \end{array}\right] -5R_1 \sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 5-5(1) & 6-5(2) & 7-5(3) & 8-5(4) \\ 9 & 10 & 11 & 12 \end{array}\right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & -4 & -8 & -12 \\ 9 & 10 & 11 & 12 \end{array}\right]$$

Matrix operations

If A is a square matrix of order m × m, and if there exists another square matrix B of the same order m × m, such that AB = BA = I, then, A is said to be invertible matrix and B is called the inverse matrix of A and it is denoted by A^{-1}.

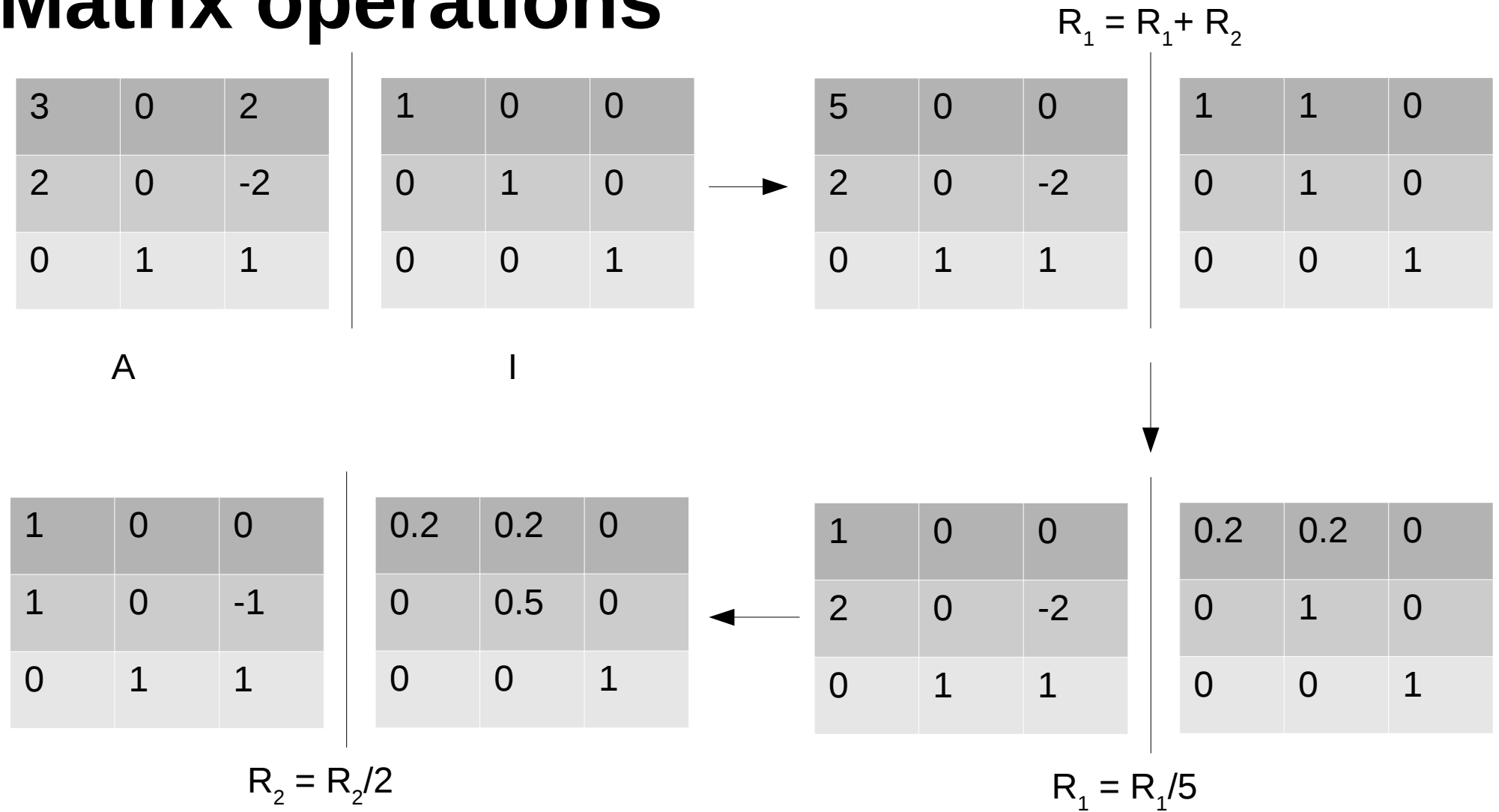
A rectangular matrix does not possess its inverse, since for the products BA and AB to be defined and to be equal, it is necessary that matrices A and B should be square matrices of the same order.

If A and B are invertible matrices of same order, then $(AB)^{-1} = B^{-1} A^{-1}$.

Inverse martix can be calculated by the matrix transformations (row and column transformations).

A = AI
I = AB via elementary row and column operations

Matrix operations



Matrix operations

1	0	0	0.2	0.2	0
0	0	-1	-0.2	0.3	0
0	1	1	0	0	1

$R_2 = R_2 - R_1$

1	0	0	0.2	0.2	0
0	1	0	-0.2	0.3	1
0	1	1	0	0	1

$R_2 = R_2 + R_3$

1	0	0	0.2	0.2	0
0	1	0	-0.2	0.3	1
0	0	1	0.2	-0.3	0

$R_3 = R_3 - R_2$

Matrix operations

Find the inverse of the given matrix

3	2	1
2	3	1
1	1	4

Matrix operations

A system of linear equations can be solved using the matrix inversion operation

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + \ldots\ldots\ldots + a_{1n}x_n = h_1$$
$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + \ldots\ldots\ldots + a_{2n}x_n = h_2$$
$$\vdots$$
$$a_{n1}x_1 + a_{n2}x_2 + a_{n3}x_3 + \ldots\ldots\ldots + a_{nn}x_n = h_n$$

$$\mathbf{A\,X = H}$$
$$\mathbf{A^{-1}AX = A^{-1}H}$$
$$\mathbf{IX = A^{-1}H}$$
$$\mathbf{X = A^{-1}H}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \ldots\ldots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \ldots\ldots & a_{2n} \\ \vdots & & & & \\ a_{n1} & a_{n2} & a_{n3} & \ldots\ldots & a_{nn} \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \mathbf{H} = \begin{bmatrix} h_1 \\ h_2 \\ \vdots \\ h_n \end{bmatrix}$$

Matrix operations

Solve the following using matrix method

$2x_1 + x_2 + 2x_3 + x_4 = 6$

$6x_1 - 6x_2 + 6x_3 + 12x_4 = 36$

$4x_1 + 3x_2 + 3x_3 - 3x_4 = -1$

$2x_1 + 2x_2 - x_3 + x_4 = 10$

Matrix operations

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \ldots\ldots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \ldots\ldots & a_{2n} \\ . & & & & \\ . & & & & \\ a_{n1} & a_{n2} & a_{n3} & \ldots\ldots & a_{nn} \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ . \\ . \\ x_n \end{bmatrix} \quad \mathbf{H} = \begin{bmatrix} h_1 \\ h_2 \\ . \\ . \\ h_n \end{bmatrix}$$

$|\mathbf{A} - \lambda\mathbf{I}| = 0$, here λ is scalar

λ is the eigen value of **A**

Matrix operations

$$\det(A) = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix}$$

$$= a(ei - hf) - d(bi - hc) + g(bf - ec)$$

$$\begin{vmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{vmatrix} = |A| = a \cdot \begin{vmatrix} f & g & h \\ j & k & l \\ n & o & p \end{vmatrix} - b \cdot \begin{vmatrix} e & g & h \\ i & k & l \\ m & o & p \end{vmatrix} + c \cdot \begin{vmatrix} e & f & h \\ i & j & l \\ m & n & p \end{vmatrix} - d \cdot \begin{vmatrix} e & f & g \\ i & j & k \\ m & n & o \end{vmatrix}$$

Matrix operations

3	-3	3
2	1	1
1	5	6

Find the eigen values of the given matrix

Eigen values also be determined by diagonilization of matrix.

Matrix operations

Inverse of the matrix can also be calculated using $A^{-1} = \frac{adj(A)}{|A|}$

Consider a 3×3 matrix

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}.$$

Its cofactor matrix is

$$\mathbf{C} = \begin{pmatrix} +\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & +\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ -\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & +\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & -\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ +\begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} & -\begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} & +\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{pmatrix}.$$

$$\operatorname{adj}(\mathbf{A}) = \mathbf{C}^{\mathsf{T}} = \begin{pmatrix} +\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & -\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & +\begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \\ -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & +\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & -\begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} \\ +\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} & -\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} & +\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{pmatrix}$$

Matrix operations

3	-3	4
2	-3	4
0	-1	1

Find the inverse of matrix

Vectors: computer

```
import numpy as np

list1 = [1, 2, 3]
list2 = [[10],
        [20],
        [30]]

vector1 = np.array(list1)

vector2 = np.array(list2)

print("Horizontal Vector")
print(vector1)

print("----------------")

print("Vertical Vector")
print(vector2)
```

Vectors: computer

```
import numpy as np
list1 = [5, 6, 9]
list2 = [1, 2, 3]
vector1 = np.array(list1)
print("First Vector          : " + str(vector1))
vector2 = np.array(list2)
print("Second Vector         : " + str(vector2))
addition = vector1 + vector2
print("Vector Addition       : " + str(addition))
```

Vectors: computer

```
import numpy as np
list1 = [5, 6, 9]
list2 = [1, 2, 3]
vector1 = np.array(list1)
print("First Vector          : " + str(vector1))
vector2 = np.array(list2)
print("Second Vector         : " + str(vector2))
addition = vector1 + vector2
print("Vector Addition       : " + str(addition))
subtraction = vector1 - vector2
print("Vector Substraction   : " + str(subtraction))
```

Vectors: computer

```
import numpy as np
list1 = [5, 6, 9]
list2 = [1, 2, 3]
vector1 = np.array(list1)
print("First Vector          : " + str(vector1))
vector2 = np.array(list2)
print("Second Vector         : " + str(vector2))
addition = vector1 + vector2
print("Vector Addition       : " + str(addition))
subtraction = vector1 - vector2
print("Vector Substraction   : " + str(subtraction))
multiplication = vector1 * vector2
print("Vector Multiplication : " + str(multiplication))
division = vector1 / vector2
print("Vector Division       : " + str(division))
```

Vectors: computer

```
import numpy as np
list1 = [5, 6, 9]
list2 = [1, 2, 3]
vector1 = np.array(list1)
print("First Vector  : " + str(vector1))
vector2 = np.array(list2)
print("Second Vector : " + str(vector2))
dot_product = vector1.dot(vector2)
print("Dot Product   : " + str(dot_product))
```

Vectors: computer

```
import numpy
import array

a=[5,6,9]
b=[1,3,4]

dot = 0.0;

for i in range(len(a)):
      dot += a[i] * b[i]

print("dot_product = "+ str(dot))
```

Vectors: computer

```
Import numpy as np
A =[2, 7, 4]
B =[3, 9, 8]

vector1=np.array(A)
vector2=np.array(B)
cross_product=np.cross(vector1, vector2)
print("Cross product :" +str(cross_product))
```

Vectors: computer

```
vect_A = [3, -5, 4]
vect_B = [2, 6, 5]
cross_P = [ ]

cross_P.append(vect_A[1] * vect_B[2] - vect_A[2] * vect_B[1])
cross_P.append(vect_A[2] * vect_B[0] - vect_A[0] * vect_B[2])
cross_P.append(vect_A[0] * vect_B[1] - vect_A[1] * vect_B[0])

print("Cross product:", end =" ")
for i in range(0, 3):
        print(cross_P[i], end =" ")
```

Matrices: computer

```
import numpy as np
M1 = np.array([[5, -10, 15],
[3, -6, 9], [-4, 8, 12]])
print(M1)
```

Matrices: computer

```
import numpy as np
M1 = np.array([[3, 6, 9], [5, -10, 15], [-7, 14, 21]])
M2 = np.array([[9, -18, 27], [11, 22, 33], [13, -26, 39]])
M3 = M1 + M2
print(M3)
```

Matrices: computer

```
import numpy as np
M1 = np.array([[3, 6, 9], [5, -10, 15], [-7, 14, 21]])
M2 = np.array([[9, -18, 27], [11, 22, 33], [13, -26, 39]])
M3 = M1 - M2
print(M3)
```

Matrices: computer

```
import numpy as np
M1 = np.array([[3,6,9],[5,-10,15],[-7,14,21]])
M2 = np.array([[9,-18,27],[11,22,33],[13,-26,39]])
M3 = M1.dot(M2)
print(M3)
```

```
import numpy as np
mat1 = ([1, 6, 5],[3 ,4, 8],[2, 12, 3])
Mat2 = ([3, 4, 6],[5, 6, 7],[6, 56, 7])
res = np.dot(mat1,mat2)
print(res)
```

Matrices: computer

```
matrix1 = [[12,7,3],
        [4 ,5,6],
        [7 ,8,9]]
matrix2 = [[5,8,1],
        [6,7,3],
        [4,5,9]]

res = [[0 for x in range(3)] for y in range(3)]

for i in range(len(matrix1)):
    for j in range(len(matrix2[0])):
        for k in range(len(matrix2)):

            res[i][j] += matrix1[i][k] * matrix2[k][j]

print (res)
```

Matrices: computer

```
M1 = np.array([[3, 6, 9], [5, -10, 15], [4,8,12]])
M2 = M1.transpose()
print(M1)
print(M2)
```

Matrices: computer

```
X = [[12,7],
     [4 ,5],
     [3 ,8]]

result = [[0,0,0],
          [0,0,0]]


for i in range(len(X)):

   for j in range(len(X[0])):
       result[j][i] = X[i][j]

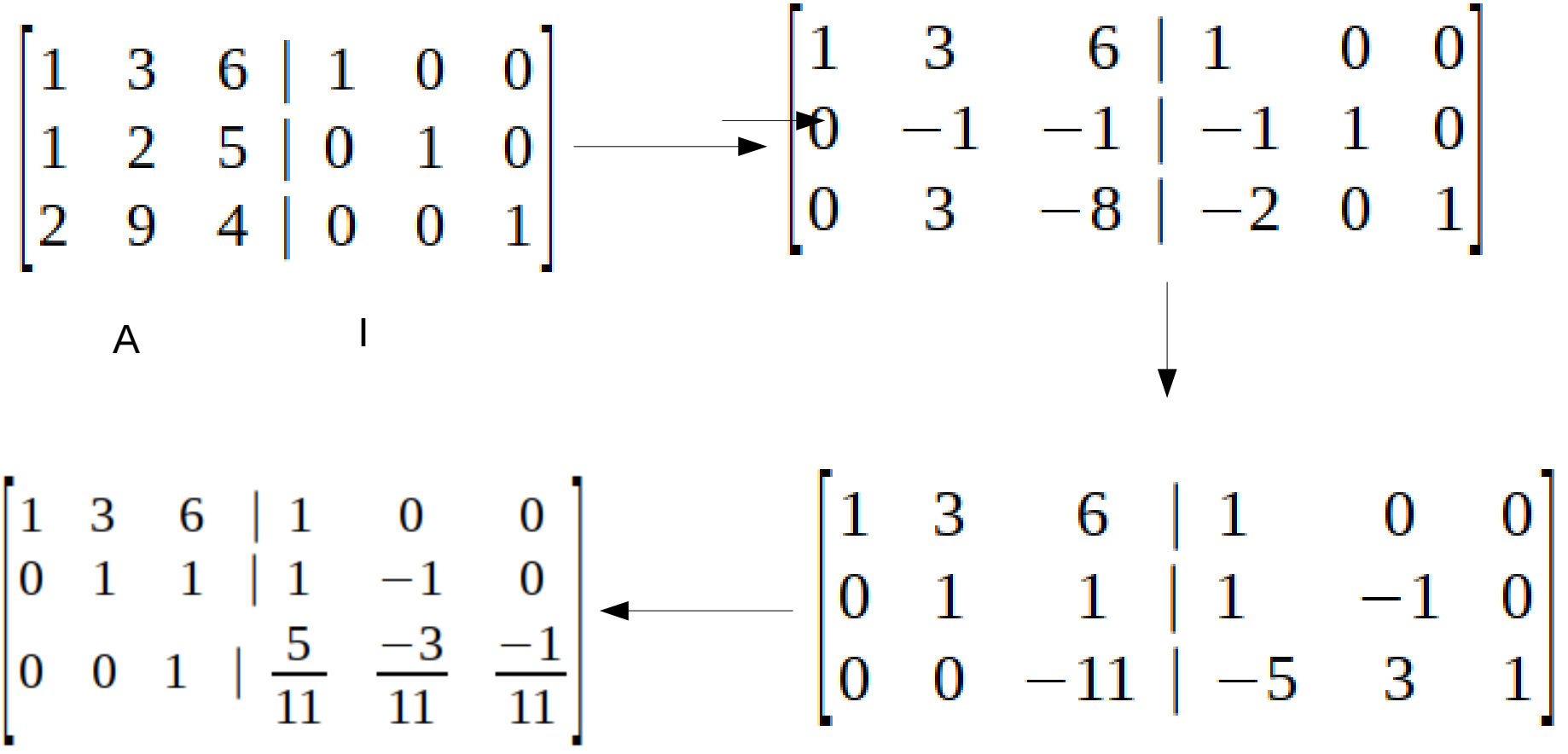
for r in result:
   print(r)
```

Matrices: computer

```
import numpy as np

A = np.array([[6, 1, 1], [4, -2, 5], [2, 8, 7]])
B = np.linalg.inv(A)
print (B)
```

Matrices: computer



Matrices: computer

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 6 & 1 & 0 & 0 \\ 0 & 1 & 0 & \frac{6}{11} & \frac{-8}{11} & \frac{1}{11} \\ 0 & 0 & 1 & \frac{5}{11} & \frac{-3}{11} & \frac{-1}{11} \end{array}\right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 6 & \frac{-7}{11} & \frac{24}{11} & \frac{-3}{11} \\ 0 & 1 & 0 & \frac{6}{11} & \frac{-8}{11} & \frac{1}{11} \\ 0 & 0 & 1 & \frac{5}{11} & \frac{-3}{11} & \frac{-1}{11} \end{array}\right]$$

$$\downarrow$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{-37}{11} & \frac{42}{11} & \frac{3}{11} \\ 0 & 1 & 0 & \frac{6}{11} & \frac{-8}{11} & \frac{1}{11} \\ 0 & 0 & 1 & \frac{5}{11} & \frac{-3}{11} & \frac{-1}{11} \end{array}\right]$$

Matrices: computer

3	3	4
2	-3	4
0	-1	1